European University of Lefke
Faculty of Engineering
Computer Engineering Department

Grade

EE431 - Digital Image Processing
2024 - 2025 Spring
Instructor: Asst. Prof. Dr. Cem Kalyoncu

Midterm
18 April 2025

ID: _ _ _ _ _ _ _ _ _ _ _ _ _ Name: _ _ _ _ _ _ _ _ _ _ _ _ _

Question 2: Write a function called `Halve` that will shrink a given image into half its size using a special interpolation technique. In this technique, you should get the average of 4 pixels and write it as the output including (0, 0). ***Hint:*** *Details are limited as it is your job to figure out how to perform this operation. Coding part is quite simple.* (ME2, ME3, LO4) **[8 points]**

Question 3: The following spatial filter window is given. (ME4, LO2, LO3)

-1	0	0
0	4	0
0	0	-1

a. What is the filter type **[1 point]**

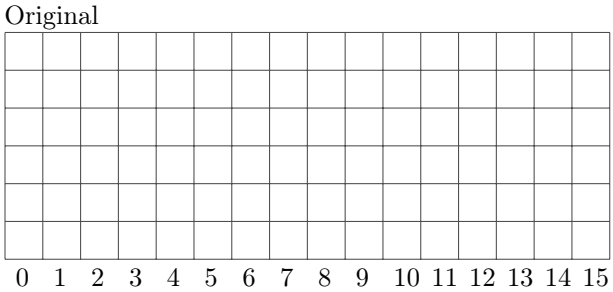
b. Apply the given filter to the following 4-bit image using mirror technique for out of bounds values. Write the result in the provided space. You should do intermediate calculations on this paper. **[6 points]**

11	10	4
11	12	5
12	4	6

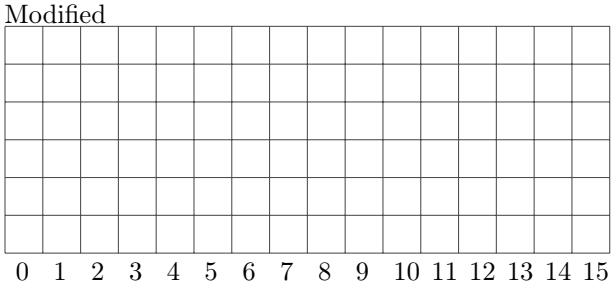
Output

c. Fill the following tables with the histograms of the original and the modified images by filling in the following bar graphs. Remember to include border pixels in the output image. **[5 points]**

Original



Modified



Question 4: Perform histogram equalize to the following 3bit grayscale image. Perform the calculations using the the table below. Write the resultant image into the given box. ***Hint:*** *To make sure that you won't making any mistakes, sum your counts (should be 25) and probabilities (should be 1).* (ME1, ME2, LO2, LO4) **[9 points]**

4	4	6	5	3
0	6	5	7	6
2	2	6	6	3
2	3	2	7	2
7	7	6	5	4

Original					**Final**
0					
1					
2					
3					
4					
5					
6					
7					

Result